

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary Winter Examination – 2024

Course: B. Tech

Branch: Civil Engineering

Semester: V

Subject Code & Name: Concrete Technology

(CV504_Y17)

Max Marks: 60

Date: 13/02/2025

Duration: 3 Hr.

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

	(CO)	Marks
Q. 1 Objective type questions. (Compulsory Question)		12
1 Which of the following is the first step in the manufacturing process of cement?	CO 1	
a) Mixing of ingredients b) Heating in a rotary kiln c) Crushing of raw materials d) Grinding of the clinker		
2 Which compound is responsible for the initial setting of cement?	CO 1	1
a) Tricalcium silicate (C3S) b) Dicalcium silicate (C2S) c) Tricalcium aluminate (C3A) d) Tetracalcium aluminoferrite (C4AF)		
3 Which of the following is a property of fine aggregates?	CO 1	
a) Angularity b) Particle size less than 4.75 mm c) Absence of clay particles d) All of the above		
4 Which of the following is a factor affecting the workability of concrete?	CO 1	1

	a) Water-cement ratio	b) Size of aggregates		
	c) Type of cement	d) All of the above		
5	Which method is used to measure the workability of concrete?		CO 1	1
	a) Slump test	b) Ve bee test		
6	c) Compaction factor test	d) All of the above	CO 1	
	What is the primary purpose of curing concrete?			
	a) To reduce the temperature of concrete			
	b) To increase the strength of concrete			
	c) To prevent segregation			
	d) To speed up setting time			
7	Which type of admixture is used to reduce the water content in concrete while maintaining workability?		CO 2	1
	a) Plasticizers	b) Retarders		
	c) Accelerators	d) Air-entraining agents		
8	What is the primary role of an air-entraining agent in concrete?		CO 2	
	a) To increase strength	b) To improve workability		
	c) To make the concrete more resistant to freeze-thaw cycles			
	d) To accelerate the setting time			
9	The characteristic strength of concrete is defined as the strength below which:		CO 3	1
	a) 95% of the cubes fail	b) 5% of the cubes fail		
	c) 10% of the cubes fail	d) No cubes fail		
10	Which of the following tests measures the tensile strength of concrete?		CO 3	
	a) Compression test	b) Flexural test		
	c) Split tensile test	d) Vee-bee test		

11	Which of the following is a type of shrinkage that occurs in concrete due to evaporation of moisture? a) Autogenous shrinkage b) Plastic shrinkage c) Drying shrinkage d) Thermal shrinkage	CO 2	1
12	What factor primarily affects the durability of concrete? a) Water-cement ratio b) Cement content c) Exposure conditions d) All of the above	CO 3	1
Q. 2	Solve the following.		12
A)	Define Aggregates. Explain Classification of aggregates.	CO 1	6
B)	Describe the manufacturing process of Ordinary Portland Cement (OPC).	CO 1	6
Q.3	Solve the following.		12
A)	What are the different methods of batching and mixing concrete?	CO 1	6
B)	Define workability. What are the factors affecting the workability of concrete?	CO 1	6
Q. 4	Solve Any Two of the following.		12
A)	What are plasticizers and super-plasticizers? How do these admixtures affect the workability and strength of concrete?	CO 2	6
B)	Define green concrete. What are its advantages of green concrete?	CO 2	6
C)	Illustrate types of admixtures.	CO 2	6
Q.5	Solve Any Two of the following.		12
A)	Define Creep. State factors affecting Creep.	CO 3	6
B)	What is alkali-aggregate reaction (AAR) in concrete? How does it affect the durability of concrete and what preventive measures can be taken?	CO 3	6

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| C) | Discuss the effects of sulphate and chloride attacks on concrete. | CO 3 | 6 |
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Q. 6	Solve Any Two of the following.		12
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| A) | Explain the methods of non-destructive testing (NDT) for concrete. | CO 3 | 6 |
| B) | Define Design Mix. State its Objectives. | CO 3 | 6 |
| C) | Explain steps involved in Concrete mix design by IS Method. | CO 3 | 6 |

*** End ***